

ELIZABETHTOWN, NC, USA

WMO: 720279

Lat: 34.602N

Lon: 78.578W

Elev: 41

StdP: 100.84

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.4	-15.0	1.0	0.5	-12.4	1.3	2.6	8.8	11.0	7.8	13.0	1.1	320	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	35.1	24.6	33.8	24.2	32.7	23.9	26.1	31.4	25.7	30.6	25.2	29.8	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	27.1	24.6	19.6	26.9	24.0	19.0	26.7	81.1	31.6	78.9	31.0	77.0	29.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	5.8	5.0	-7.9	37.0	1.8	1.1	-9.2	37.8	-10.3	38.4	-11.3	39.0	-12.6	39.8
			-9.2	27.1	1.7	0.7	-10.5	27.6	-11.5	28.0	-12.4	28.3	-13.7	28.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	7.6	8.7	12.5	17.5	21.8	25.8	27.2	26.6	23.7	18.6	12.2	9.7
	DBStd	8.19	5.79	5.11	5.22	4.25	3.50	2.60	2.01	2.15	2.89	4.25	4.73	5.39
	HDD10.0	341	119	79	35	3	0	0	0	0	0	1	29	74
	HDD18.3	1388	336	270	189	66	12	0	0	0	1	50	191	273
	CDD10.0	3156	43	44	114	227	366	476	534	515	411	267	95	64
	CDD18.3	1162	2	2	10	40	119	226	276	257	162	58	7	4
	CDH23.3	10333	4	15	94	345	986	2133	2751	2344	1225	385	37	14
	CDH26.7	4093	0	1	10	74	332	923	1241	996	436	78	3	0
Wind	WSAvg	1.7	1.9	2.1	2.3	2.3	1.8	1.6	1.6	1.2	1.0	1.3	1.4	1.7
Precipitation	PrecAvg	1248	95	78	94	82	92	125	138	158	141	89	85	78
	PrecMax	1612	184	173	182	150	167	317	293	312	444	189	245	148
	PrecMin	839	27	34	15	15	34	21	56	60	23	9	8	17
	PrecStd	191	49	38	41	38	35	68	56	64	98	55	55	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.6	27.9	30.5	34.7	36.1	36.4	36.1	33.9	31.0	26.4	25.2
		MCWB	18.1	17.6	17.1	18.7	22.5	24.3	24.6	24.6	22.6	22.4	18.5	19.9
	2%	DB	21.3	22.4	26.0	28.8	31.8	34.5	35.1	34.3	32.5	28.8	24.1	22.6
		MCWB	16.7	15.8	17.2	18.7	21.2	24.3	24.9	24.8	23.2	21.3	18.3	19.0
	5%	DB	19.1	20.0	23.8	27.3	30.1	32.9	33.8	32.9	31.1	27.3	22.2	21.0
		MCWB	15.5	14.3	15.9	18.0	21.0	23.5	24.6	24.5	22.8	20.5	17.1	18.3
	10%	DB	17.2	17.6	21.4	25.3	28.4	31.7	32.6	31.8	29.4	25.7	20.3	18.8
		MCWB	14.0	13.0	14.9	17.3	20.6	23.0	24.3	24.2	22.5	20.0	16.6	16.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	19.3	19.9	23.0	24.3	26.3	26.9	26.8	25.7	24.5	21.7	21.7
		MCDB	22.2	23.0	23.6	25.7	29.0	32.3	32.3	31.7	29.3	27.0	24.4	23.5
	2%	WB	17.9	17.5	18.9	21.6	23.2	25.7	26.2	26.0	25.1	23.6	20.3	20.2
		MCDB	20.2	20.3	23.0	25.0	27.6	31.1	31.7	30.9	28.9	26.5	22.0	21.8
	5%	WB	16.3	15.7	17.8	20.2	22.6	25.0	25.7	25.6	24.5	22.6	18.9	18.7
		MCDB	18.6	19.0	21.4	23.9	26.9	30.0	30.8	30.2	28.0	25.3	20.9	20.2
	10%	WB	14.5	13.9	16.7	19.1	21.9	24.4	25.2	25.2	23.8	21.5	17.3	16.7
		MCDB	16.8	16.9	20.1	23.2	26.2	29.1	30.0	29.4	27.1	24.1	19.4	18.6
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	12.2	12.4	11.2	10.5	10.3	9.7	9.8	10.7	11.6	10.6
		MCDBR	12.3	13.6	14.2	14.0	12.7	11.6	11.2	10.9	11.6	11.5	12.7	11.4
		MCWBR	8.5	8.4	6.7	5.9	4.2	3.7	3.4	3.4	3.8	5.2	7.5	7.9
	5% WB	MCDBR	11.1	12.5	11.7	11.1	10.2	10.6	10.2	9.7	8.8	8.7	10.1	10.3
		MCWBR	8.9	9.7	7.2	6.0	4.0	3.9	3.5	3.4	3.6	4.9	7.7	8.5
Clear-Sky Solar Irradiance	taub	0.322	0.332	0.367	0.403	0.465	0.497	0.540	0.516	0.431	0.390	0.345	0.321	
	taud	2.493	2.458	2.387	2.281	2.154	2.110	2.009	2.067	2.303	2.384	2.466	2.529	
	Ebn at Noon	886	914	905	884	829	799	762	773	829	836	848	865	
	Edh at Noon	85	97	114	132	152	159	175	162	122	104	86	78	
All-Sky Solar Radiation	RadAvg	2.71	3.37	4.45	5.57	6.07	6.33	6.05	5.41	4.61	3.85	2.98	2.38	
	RadStd	0.16	0.33	0.34	0.36	0.47	0.40	0.38	0.29	0.36	0.45	0.24	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+121

Nomenclature: See separate page